

Counting k -free ideals in arithmetic progressions in number fields

Caylee Spivey

University of Connecticut

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This project began at the sixth iteration of Rethinking Number Theory and is joint work with

- Adrian Barquero-Sanchez (University of Costa Rica)
- Jack Heimrath (Queen's University)
- Bernd Sing (University of the West Indies)
- Nicolás Sirolli (University of Buenos Aires)
- Michael Wijaya (Trinity School)

Caylee



Michael



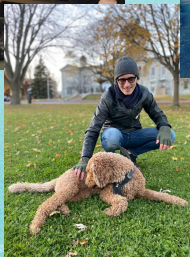
Adrián



Team
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Nicolás



Jack



Bernd

The distribution of k -free integers

For $k \geq 2$, an integer n is called k -free if it is not divisible by a k th power greater than 1.

The following is the earliest result about of k -free numbers.

Theorem (Gegenbauer, 1885)

For $k \geq 2$,

$$\#\{n \in \mathbb{Z}^+ \mid n \leq X \text{ and } n \text{ is } k\text{-free}\} = \frac{X}{\zeta(k)} + O(X^{1/k}).$$

Arithmetic progressions

A natural follow up is counting k -free numbers in an arithmetic progression $a \bmod m$ where $\gcd(a, m) = 1$. Let $Q^k(X, m, a)$ be the k -free numbers $n \leq X$ such that $n \equiv a \bmod m$. We have

$$Q^k(X, m, a) = \frac{1}{m \zeta(k)} \prod_{p|m} \left(1 - \frac{1}{p^k}\right)^{-1} X + E(X, m, a).$$

with a bound on the error term due to Prachar (1958):

$$E(X, m, a) = O \left(\left(\frac{X}{m^{\frac{1}{k}}} \right)^{\frac{1}{k}} + m^{\frac{1}{k}} k^{\omega(m)} \right)$$

where $\omega(m)$ is the number of prime factors of m . Thus

$$E(X, m, a) = O_m(\sqrt[k]{x}).$$

Counting ideals in number fields

Counting k -free numbers extends to counting k -free ideals in number fields. For a number field K of degree d , let $\mathcal{H}_K(X)$ be the number of nonzero ideals in $\mathcal{O}_K$ of norm at most X . The asymptotic growth of $\mathcal{H}_K(X)$ is given by

$$\mathcal{H}_K(X) = \rho_K X + O(X^{1-1/d}),$$

where

$$\rho_K := \operatorname{Res}_{s=1}(\zeta_K(s)) = \frac{2^{r_1}(2\pi)^{r_2} \operatorname{Reg}_K h_K}{w_K |D_K|^{1/2}}.$$

Generalized arithmetic progressions

Arithmetic progressions $\{n \equiv a \bmod m\}$ where $\gcd(a, m) = 1$ generalize in a number field K by using *ray class groups*.

Let $\mathfrak{m} = \mathfrak{m}_0 \mathfrak{m}_\infty$ be a modulus for K where $\mathfrak{m}_0$ is the finite part and $\mathfrak{m}_\infty$ is the infinite part. Denote the *ray class group of K modulo $\mathfrak{m}$* as $\text{Cl}_K^\mathfrak{m}$, so $\text{Cl}_\mathbb{Q}^{(m)\infty} \cong (\mathbb{Z}/m)^\times$.

Counting ideals in ray classes

Let K be a number field of degree d and $\mathfrak{m}$ be a fixed modulus for K . Suppose that $k \geq 2$ and $\mathfrak{A} \in \text{Cl}_K^{\mathfrak{m}}$ is a ray class modulo $\mathfrak{m}$. Let $\mathcal{H}_K^{\mathfrak{m}}(X; \mathfrak{A})$ be the number of nonzero ideals of $\mathcal{O}_K$ in $\mathfrak{A}$ of norm at most X that are relatively prime to $\mathfrak{m}_0$.

As with $\mathcal{H}_K(X)$, we have an asymptotic formula for $\mathcal{H}_K^{\mathfrak{m}}(X; \mathfrak{A})$. For any ray class $\mathfrak{A} \in \text{Cl}_K^{\mathfrak{m}}$, as $X \rightarrow \infty$

$$\mathcal{H}_K^{\mathfrak{m}}(X; \mathfrak{A}) = \rho_K^{\mathfrak{m}} X + O(X^{1-1/d}),$$

where

$$\rho_K^{\mathfrak{m}} = \text{Res}_{s=1}(\zeta_K(s, \mathfrak{A}))$$

for

$$\zeta_K(s, \mathfrak{A}) := \sum_{\substack{\mathfrak{a} \in \mathfrak{A} \\ \mathfrak{a} \subset \mathcal{O}_K}} \frac{1}{N(\mathfrak{a})^s} \quad \text{for } \text{Re}(s) > 1.$$

Note: The residue $\rho_K^{\mathfrak{m}}$ is *the same* for all $\mathfrak{A}$ in $\text{Cl}_K^{\mathfrak{m}}$.

Counting functions of ideals of bounded norm

For $k \geq 2$, a nonzero ideal $\mathfrak{a}$ in $\mathcal{O}_K$ is called k -free if it is not divisible by the k th power of an ideal other than (1). We now define the counting functions that are our main interest.

Definition

Let K be a number field and $\mathfrak{m}$ be a fixed modulus for K . Also, suppose $k \geq 2$, and let $\mathfrak{A} \in \text{Cl}_K^{\mathfrak{m}}$ be a ray class modulo $\mathfrak{m}$. Then for any $X > 1$ we define the following functions.

- (i) Let $Q_K^k(X)$ denote the number of k -free ideals $\mathfrak{a}$ in $\mathcal{O}_K$ with $N(\mathfrak{a}) \leq X$.
- (ii) Let $Q_K^k(X; \mathfrak{A})$ denote the number of k -free ideals $\mathfrak{a}$ in $\mathcal{O}_K$ and $\mathfrak{A}$ with $N(\mathfrak{a}) \leq X$.

Theorem 1 (BHSSSW)

Let K be a number field of degree d and $\mathfrak{m}$ be a modulus for K . Also, let $k \geq 2$ and $\mathfrak{A} \in \text{Cl}_K^{\mathfrak{m}}$ be a fixed ray class modulo $\mathfrak{m}$. Then for $X > 1$,

$$Q_K^k(X; \mathfrak{A}) = \frac{\rho_K^{\mathfrak{m}}}{\zeta_K(k)} \prod_{\mathfrak{p}|\mathfrak{m}_0} \left(1 - \frac{1}{N(\mathfrak{p})^k}\right)^{-1} X + E_{K,\mathfrak{m}}^k(X),$$

where $\rho_K^{\mathfrak{m}} = \text{Res}_{s=1}(\zeta_K(s; \mathfrak{A}))$ and

$$E_{K,\mathfrak{m}}^k(X) = \begin{cases} O_{\mathfrak{m}}(X^{1-1/d} \log X), & \text{if } k = \frac{2d-1}{d-1}, \\ O_{\mathfrak{m}}(X^{(2-1/d)/k}), & \text{if } k < \frac{2d-1}{d-1}, \\ O_{\mathfrak{m}}(X^{1-1/d}), & \text{if } k > \frac{2d-1}{d-1}. \end{cases}$$

If $K = \mathbb{Q}$ and $\mathfrak{m} = (m)_{\infty}$, only the second case occurs ($k < \infty$) and it says

$$E_{\mathbb{Q},(m)_{\infty}}^k(X) = O_m(X^{1/k}).$$

Counting tuples of ideals of bounded norm

It is also of interest to count *tuples* of ideals of bounded norm.

Definition

Let K be a number field and $\mathfrak{m}$ be a fixed modulus for K . Also, suppose that $k, r \in \mathbb{Z}^+$, and let $\mathfrak{A} \in \text{Cl}_K^{\mathfrak{m}}$ be a ray class modulo $\mathfrak{m}$. Then for any $X > 1$ we define the following functions.

- (i) Let $Q_K^{k,r}(X)$ be the number of r -tuples $(\mathfrak{a}_1, \dots, \mathfrak{a}_r)$ of nonzero ideals $\mathfrak{a}_i$ in $\mathcal{O}_K$ with $N(\mathfrak{a}_i) \leq X$ that have no common k th power factor besides (1) (“relatively k -prime”).
- (ii) Let $Q_K^{k,r}(X; \mathfrak{A})$ be the number of r -tuples $(\mathfrak{a}_1, \dots, \mathfrak{a}_r)$ of nonzero ideals $\mathfrak{a}_i$ in $\mathfrak{A}$ with $N(\mathfrak{a}_i) \leq X$ such that the ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_r$ are relatively k -prime.

Counting tuples of ideals of bounded norm

Here is a recent result about counting tuples of ideals.

Theorem (Sittinger, 2010)

Let K be a number field. Fix $k, r \in \mathbb{Z}^+$ not both equal to 1. Let $\mathcal{H}_K(X)$ be the number of ideals in $\mathcal{O}_K$ with norm at most X , and $Q_K^{k,r}(X)$ be the number of ordered r -tuples of ideals in $\mathcal{O}_K$ with norm at most X that are relatively k -prime. Then

$$\lim_{X \rightarrow \infty} \frac{Q_K^{k,r}(X)}{\mathcal{H}_K(X)} = \frac{1}{\zeta_K(kr)}.$$

Theorem 2 (BHSSSW)

Let K be a number field of degree d and let $\mathfrak{m}$ be a modulus for K . Let $k, r \in \mathbb{Z}^+$ not both equal 1 and let $\mathfrak{A} \in \text{Cl}_K^{\mathfrak{m}}$ be a fixed ray class modulo $\mathfrak{m}$. Then for $X > 1$ we have

$$Q_K^{k,r}(X; \mathfrak{A}) = \frac{(\rho_K^{\mathfrak{m}})^r}{\zeta_K(kr)} \prod_{\mathfrak{p}|\mathfrak{m}_0} \left(1 - \frac{1}{N(\mathfrak{p}^{kr})}\right)^{-1} X^r + E_{K,\mathfrak{m}}^{k,r}(X),$$

where $\rho_K^{\mathfrak{m}} = \text{Res}_{s=1}(\zeta_K(s; \mathfrak{A}))$ and

$$E_{K,\mathfrak{m}}^{k,r}(X) = \begin{cases} O_{\mathfrak{m}}(X^{r-1/d} \log X), & \text{if } k = \frac{2d-1}{dr-1}, \\ O_{\mathfrak{m}}(X^{(2-1/d)/k}), & \text{if } k < \frac{2d-1}{dr-1}, \\ O_{\mathfrak{m}}(X^{r-1/d}), & \text{if } k > \frac{2d-1}{dr-1}. \end{cases}$$

Question: How does the error term depend on $\mathfrak{m}$?

Here are the discussion topics from RNT 6:

- Activism and organization in academia
- Rethinking the mathematical profession: adapting to a changing world
- Building and supporting research collaborations

Questions?

Let K be a number field of degree d and $\mathfrak{m}$ be a modulus for K . Then we let

$$\zeta_K(s, \mathfrak{m}) := \sum_{\mathfrak{a} \in \text{Cl}_K^{\mathfrak{m}}} \zeta_K(s, \mathfrak{a}) \sum_{\substack{\mathfrak{a} \subset \mathcal{O}_K \\ \gcd(\mathfrak{a}, \mathfrak{m}_0) = 1}} \frac{1}{N(\mathfrak{a})^s} \quad \text{for } \text{Re}(s) > 1.$$

We observe that

$$\zeta_K(s, \mathfrak{m}) = \prod_{\mathfrak{p} \nmid \mathfrak{m}_0} \left(1 - \frac{1}{N(\mathfrak{p})^s} \right)^{-1} = \zeta_K(s) \prod_{\mathfrak{p} | \mathfrak{m}_0} \left(1 - \frac{1}{N(\mathfrak{p})^s} \right),$$

so $\zeta_K(s, \mathfrak{m})$ differs from $\zeta_K(s)$ by a finite Euler product that corresponds to the prime ideals that divide the finite part $\mathfrak{m}_0$.